# Questionnaire Based Analysis of Anxiety and Personality in a Bilateral Plantar Pressure Gait Classifier

Md Tarik Anvar
22190503033

Nishan Das
22190503037

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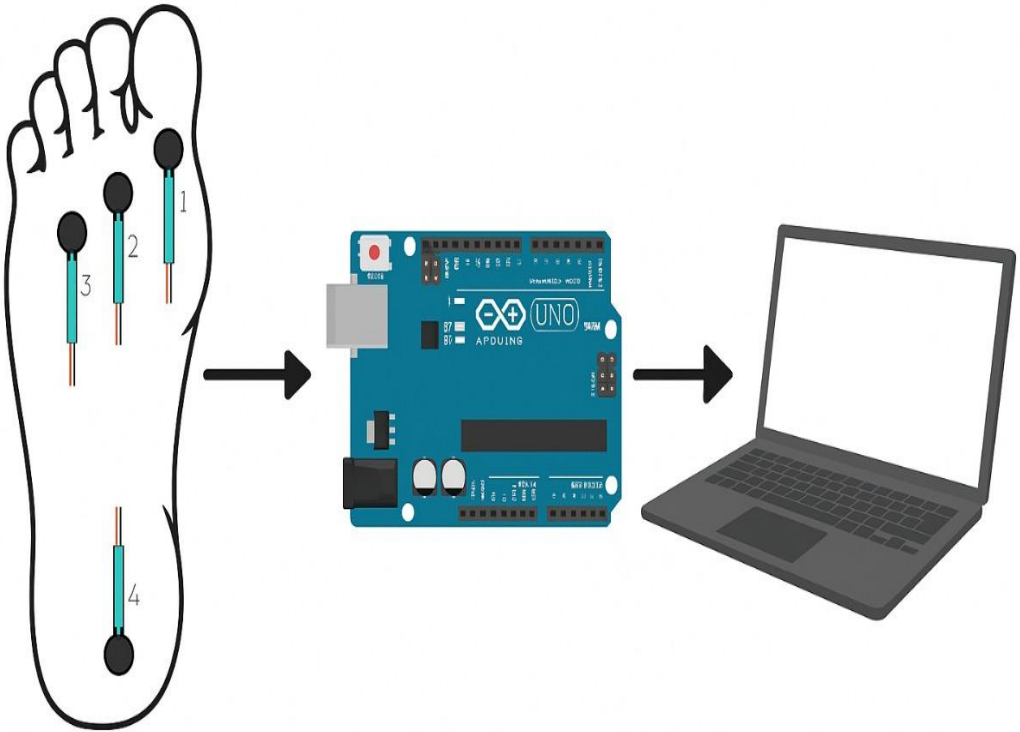
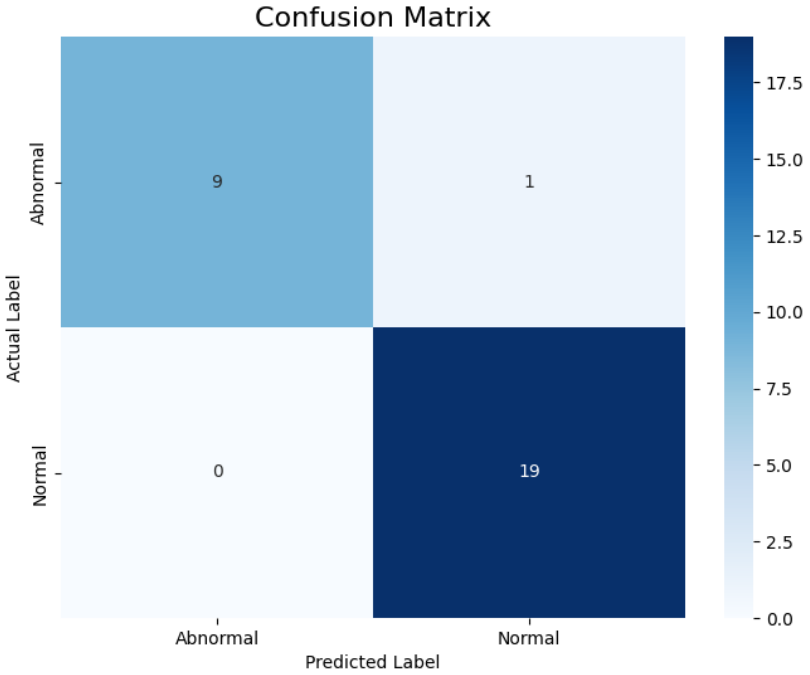
# Background and Motivation

- Developed a low-cost bilateral plantar-pressure gait classifier
- Used FSR-based wearable sensing
- Classified gait into Normal vs Abnormal
- Motivation: affordable alternative to expensive gait labs



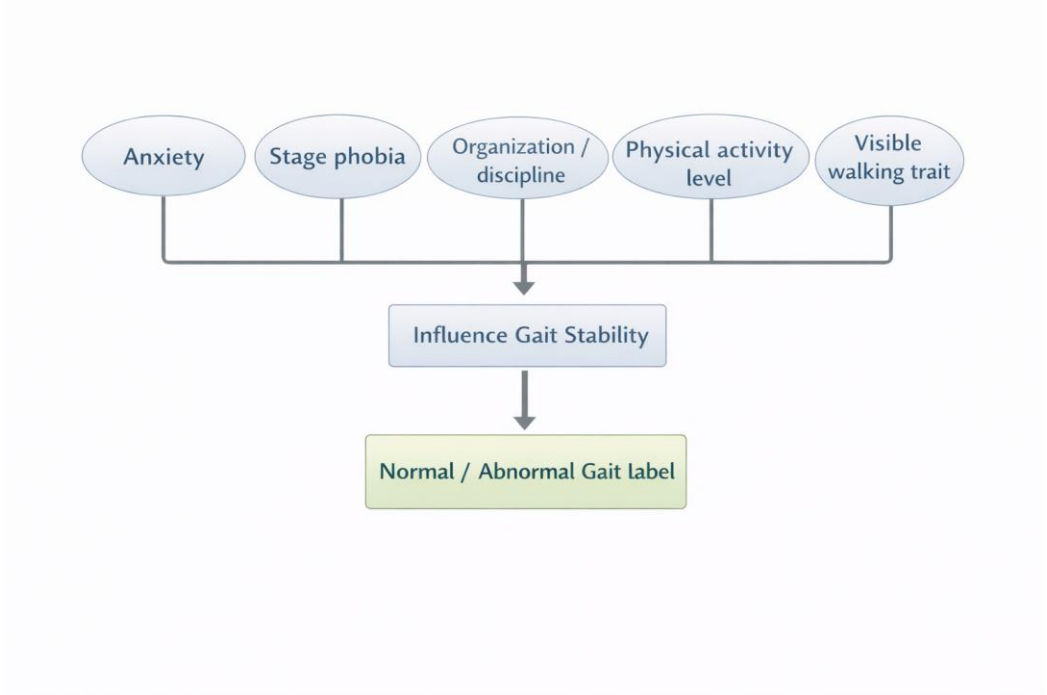
# System and Results

- Four FSRs per foot, Arduino Uno, bilateral data logging
- Feature extraction: cadence, gait asymmetry, step-time variability
- Random Forest classifier used
- Strong performance:
  1. Accuracy: 96.55%
  2. Macro-F1: 0.96
  3. ROC-AUC: 0.95



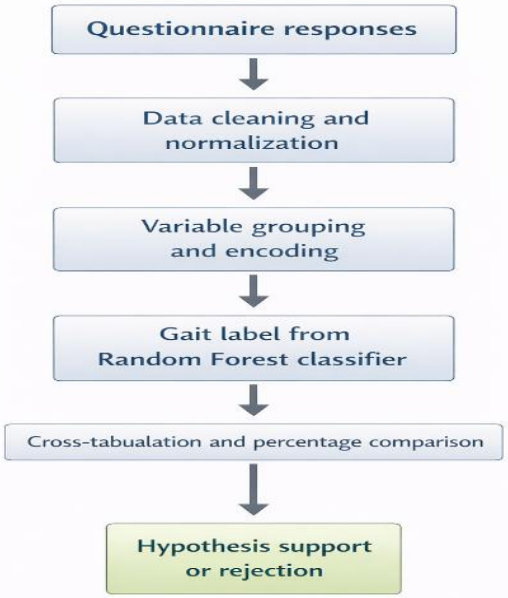
# Questionnarie-Based Behavioral Analysis

- Gait is not only biomechanical
- Psychological and behavioral factors may also influence walking style
- Extension aim: link questionnaire-based human factors with gait abnormality
- Move from only sensor-based classification to behavioral interpretation



# Questionnaire and Behavioral Data

- Questionnaire responses for:
  1. anxiety
  2. stage phobia
  3. organization and discipline
  4. physical activity level
  5. visible walking trait
  6. crowd behavior
- 97 participants
- Purpose: identify patterns related to gait

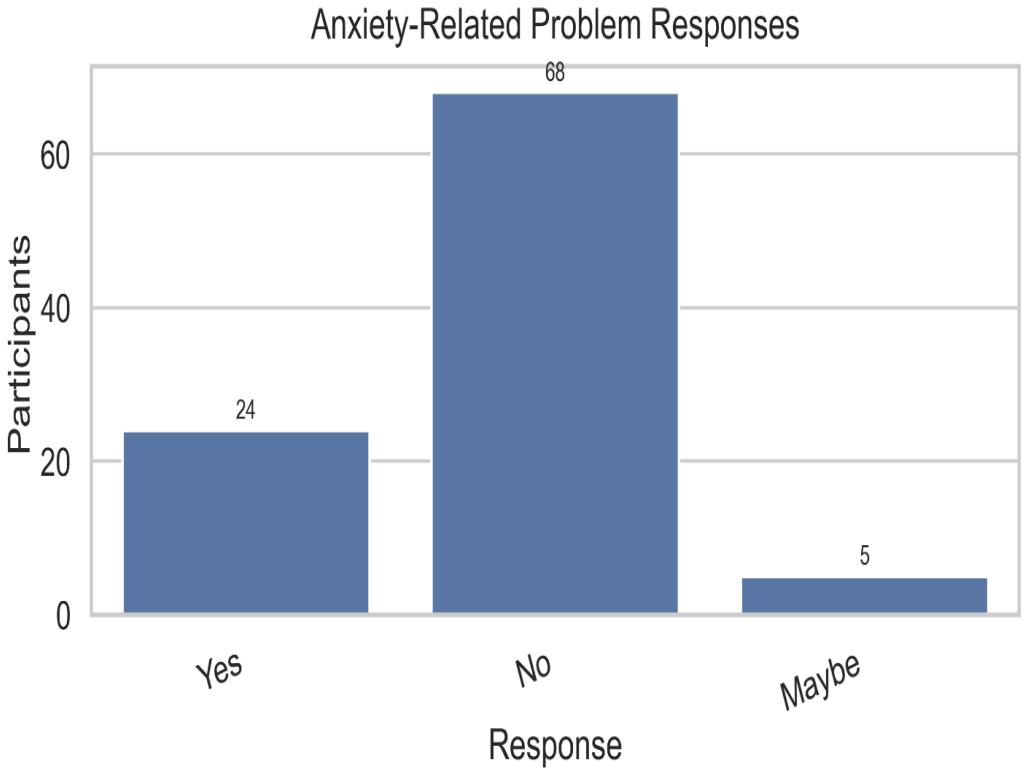


# Hypothesis Framework

- **The extended study is based on hypothesis testing to examine whether questionnaire-based behavioral factors are associated with gait status.**
- **Each hypothesis links a specific questionnaire response or grouped questionnaire variable to the gait label generated by the existing classifier.**
- **The objective is to determine whether psychological and behavioral traits such as anxiety, stage phobia, discipline, physical activity, and visible walking style show meaningful relationships with normal or abnormal gait.**
- **In the following slides, each hypothesis is presented and discussed separately, with supporting figures and interpretation.**

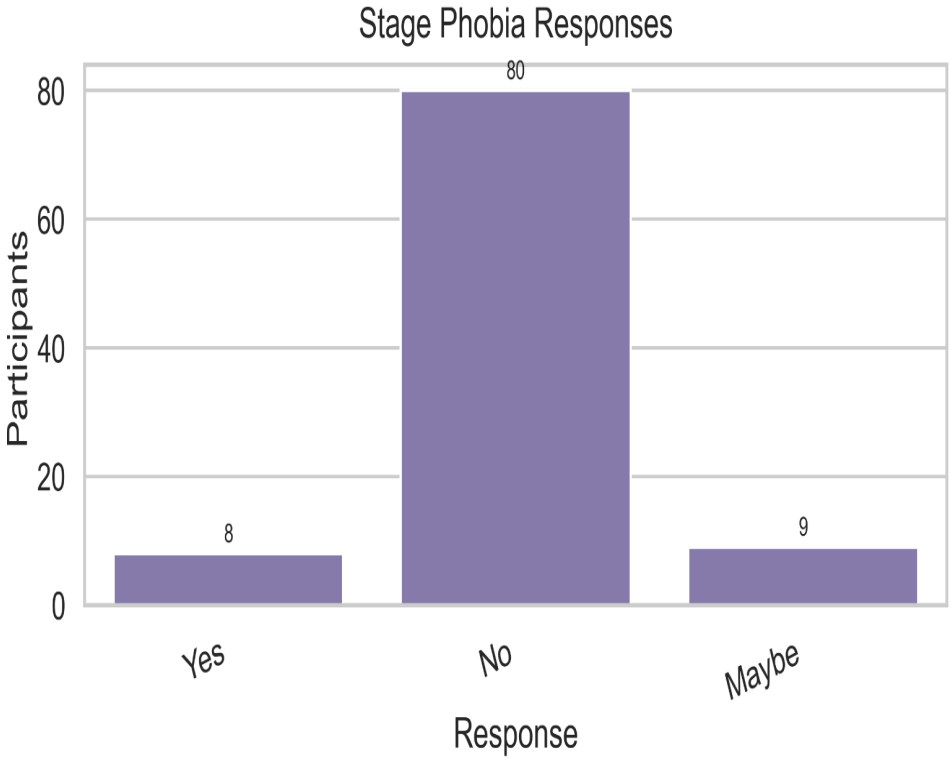
# H1 Anxiety and Abnormal Gait

- **H1: Anxiety-related problems are associated with abnormal gait**
- **Why: anxiety may increase tension, stiffness, irregular step timing**
- **Expected outcome: higher abnormal proportion in anxious participants**



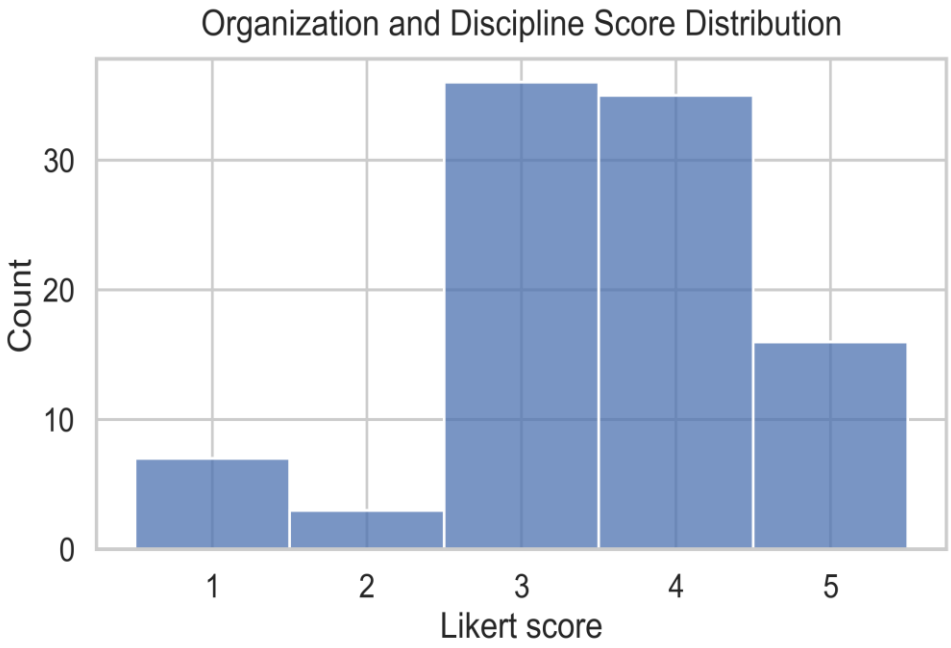
# H2 Stage Phobia and Abnormal Gait

- **H2: Stage phobia is associated with abnormal gait**
- **Why: fear of being observed may affect natural walking style**
- **Expected outcome: stage-phobia participants may show more abnormal gait**



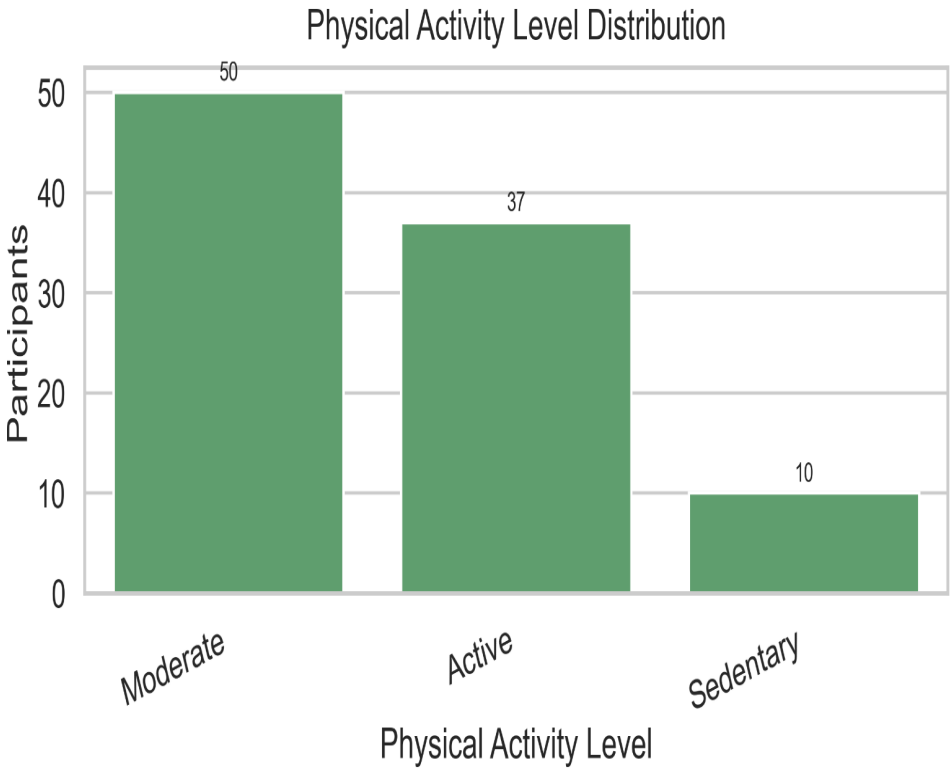
# H3 Organization / Discipline and Normal Gait

- **H3: Higher organization and discipline scores are linked with normal gait**
- **Why: disciplined behavior may reflect stable habits and more regular movement**
- **Expected outcome: score ≥ 3 more likely to correspond to normal gait**



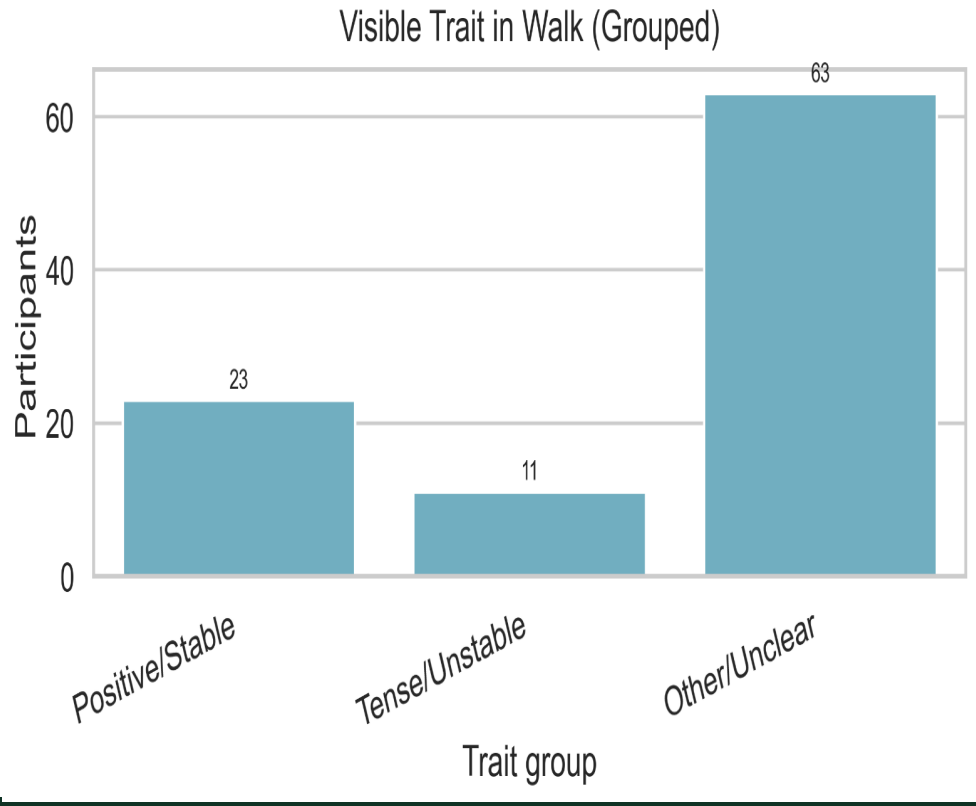
# H4 Physical Activity Level and Gait Quality

- **H4: Sedentary participants are more likely to show abnormal gait**
- **Why: physical activity supports strength, balance, and motor control**
- **Expected outcome: sedentary group may show more gait instability**



# H5 Visible Trait in Walk and Gait Stability

- **H4: Sedentary participants are more likely to show abnormal gait**
- **Why: physical activity supports strength, balance, and motor control**
- **Expected outcome: sedentary group may show more gait instability**



# Summary

- **Questionnaire provides behavioral context for gait analysis**
- **Hypotheses link psychological traits to gait labels**
- **Figures and distributions show that the dataset has enough variation for analysis**
- **Extension adds interpretability to the original classifier**

| Hypothesis | Main Questionnaire Factor | Expected Relationship |
|---|---|---|
| H1 | Anxiety-related problem | Anxiety may be linked to abnormal gait |
| H2 | Stage phobia | Stage phobia may be linked to abnormal gait |
| H3 | Organization and discipline | Higher scores may be linked to normal gait |
| H4 | Physical activity level | Sedentary participants may show abnormal gait |
| H5 | Visible trait in walk | Positive/stable traits may be linked to normal gait |

# Conclusion and Future Work

- **Project 1: strong gait classification using wearable FSRs and Random Forest**
- **Project 2: hypothesis-based behavioral extension using questionnaire data**
- **Future:**
    1. **merged statistical testing**
    2. **larger dataset**
    3. **multi-class gait classification**
    4. **sensor fusion with IMU**

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# Thank You